

**Proceedings
of the 2025 International
Conference
on Systems and Technologies
of the Digital HealthCare
(STDH – 2025)**

**2 – 6 June, 2025
Saint-Petersburg, Russia**

Technology of Localization Pathological Accumulation Foci of Radiopharmaceutical Based on Deep Learning Methods

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Abstract— This work is dedicated to the development of a technology for localizing foci of pathological accumulation of radiopharmaceuticals based on deep learning methods. The presented technology can be successfully applied in nuclear medicine and radionuclide diagnostics departments as a second opinion tool when radiologists describe radionuclide studies. Its use is particularly relevant for the early diagnosis of tumors and metastatic lesions, as well as for evaluating the effectiveness of ongoing treatment and differential diagnosis of diseases. The study examined three main approaches to analyzing scintigraphic images: neural network analysis, morphological analysis, and texture analysis. It was found that methods based on convolutional neural networks (CNN) demonstrate the highest accuracy and are the most promising for automating the process of analyzing medical images. To solve the image classification problem, modern CNN architectures were utilized and fine-tuned, including InceptionV3, DenseNet121, and ResNet101V2. The training of the models was carried out on a data set provided by specialists of the Medical City Medical Center in Tyumen, which included 68 scintigraphy images, of which 18 contained foci of pathological accumulation of a radiopharmaceutical drug. The results of the computational experiment showed that the ResNet101V2 architecture demonstrated the best performance among the considered models, achieving an accuracy of 89%, a recall of 67%, and an area under the ROC curve of 0.79. The obtained results confirm the potential of the proposed technology to improve diagnostic accuracy and minimize the influence of human error in the interpretation of scintigraphic images.

Keywords — convolutional neural networks, deep learning, machine learning, scintigraphy, googlenet, resnet, densenet

I. INTRODUCTION

Bone scintigraphy using ^{99m}Tc -MDP is one of the most commonly used clinical tools for screening bone metastases [1]. When the primary tumor infiltrates bone tissue, one or more areas of increased radionuclide uptake begin to be observed on single-photon emission computed tomography images (Figure 1). Compared to positron emission tomography (PET), bone scanning with ^{99m}Tc -MDP SPECT is more affordable and acceptable for the examination of skeletal metastases because of its high sensitivity. The ^{99m}Tc -MDP SPECT image consists of two parts: front view; back view.

However, ^{99m}Tc -MDP SPECT images suffer from low specificity. Low specificity, combined with normal absorption variation and technical artifacts, often leads to

misinterpretation of results by specialists when they manually diagnose bone metastases. It is obvious from the above mentioned that automation of the process of localization of areas of interest in bone scintigraphy images allows to minimize the human factor as well as to increase the accuracy of diagnosis.

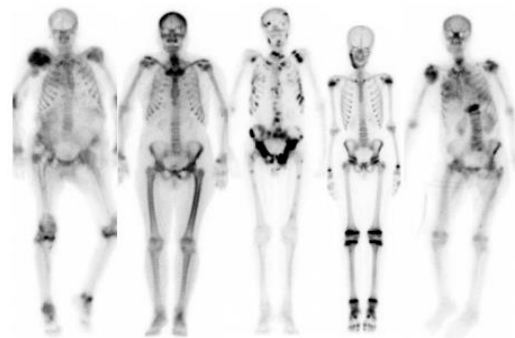


Fig. 1. Examples of scintigraphy images

Due to the use of CNN it is possible to increase the speed and efficiency of processing a large amount of data [2]. In this regard, it is important to create a technology for automatic process of analyzing bone scintigraphy images. The research objectives include literature analysis, data preprocessing, implementation of the selected localization algorithm, and computational experiment.

As a result of analyzing scientific articles on this topic, three popular classes of methods have been identified that are actively used for solving problems of medical image analysis, in particular, for detecting metastases and other pathologies: neural network approach, morphological analysis and texture analysis. Each of these methods has its own features, advantages and applications.

In neural network approach [2, 3, 4, 5], three innovative methods have been used for feature aggregation [2]. Squeeze-and-Excitation (SE): based on the analysis of dependencies between channels, the method allows to enhance the relevant features and suppress irrelevant data. The mechanism of SE operation includes two stages: first, the information is squeezed (squeeze) across channels, and then the most important channels are excited (excitation). This allows the network to adaptively redistribute attention between channels; Spatial Pyramid Pooling (SPP): Based on processing feature

maps using multilevel pooling (data dimensionality reduction), it helps extract features at different scales and resolution levels; Attention-Augmented hybridized (spatial and channel-based) attention mechanism. The spatial attention mechanism is often implemented through additional layers that analyze feature maps and generate a matrix of weights. These weights are then applied to the original feature maps to reinforce important areas. The channel attention mechanism is realized through Squeeze-and-Excitation.

Residual connections prevent gradient fading and improve information transfer between layers [1]. Residual connections are computed by adding input data to the output data of one or more layers of a neural network. This allows the network to pass information directly through the layers, bypassing possible distortions or losses. Class Activation Maps (CAMs) help to highlight significant regions of the image associated with metastases [2].

The process includes the following steps:

1. In the last convolutional layer of the network, feature maps are extracted which contain spatial information about the image.
2. For each class, weights of neurons on the fully connected layer are calculated, which show how important each feature channel is for that class.
3. The feature maps are multiplied by the corresponding weights and summed to produce a heat map that highlights the significant regions of the image.

Region Proposal Network (RPN) is a component of the Faster R-CNN architecture that generates region proposals (region proposals) potentially containing objects. For each region, a pixel-level mask is predicted. Binary cross-entropy is applied to train the masking network. Feature maps obtained from convolutional layers of the underlying network (e.g., ResNet or VGG) are fed to the RPN input. At each sliding window of the feature map, a small convolutional layer is applied that simultaneously predicts: the objectness score, which indicates the probability of an object being present in a given area; and the coordinates of the bounding box, which specifies the position and size of the area. The anchors used are a set of predefined bounding boxes of different sizes and aspect ratios that serve as templates for generating area suggestions. Area suggestions are filtered based on objectness score, and the most likely candidates are selected using Non-Maximum Suppression (NMS) to eliminate overlapping areas.

In morphological analysis [10, 11, 12, 13], segmentation includes pixel classification, Dice classification layer and focal loss [10]. Salp Swarm algorithm is used to optimize the thresholding based on sharpness index. It is a method inspired by the behavior of salps (marine organisms) that form chains to move collectively in the ocean. An initial population of salps is created, where each salp represents a possible solution to the problem. The positions of salps are randomly distributed in the search space. In a chain of salps, the first salp is considered the leader and the others follow. The leader updates its position based on the target function, aiming for the best solution.

Markov Random Field (MRF) method is applied for automatic edge segmentation [14]. MRF considers dependencies between neighboring elements and is applied to segmentation, image reconstruction and classification tasks.

Ema Rachmawati et al. proposed an approach based on Constrained Local Model (CLM), for the analysis of shape and texture variations, the principal component analysis (PCA)

was used to select the most significant features from the data matrix [15]. The principal component method is based on finding linearly uncorrelated variables called principal components. These components form a new basis in the feature space, excluding unnecessary information and reducing the dimensionality. The constrained local model is based on a statistical shape model and local detectors (also called patch experts). The general CLM approach includes the following steps:

1. Selecting a region in the image around the current estimate and projecting it into a reference frame.
2. Generating an “inverse image” for each point, where the cost of placing a point at each pixel is specified.
3. Finding a combination of points that optimizes the total cost by manipulating the shape model parameters.

The model fitting process was performed using the regularized mean shift (RLMS) method, which ensured that the model was accurately placed in the image, taking into account variations in pixel shape and intensity. Regularization allowed to increase the robustness of the method to noise and image artifacts.

In texture analysis we used median filter for noise reduction, object-oriented classification, image segmentation based on boundary analysis together with Full Lambda Scheduling (FLS) algorithm [16]. The object-oriented classification method is based on image segmentation into several homogeneous groups of pixels (objects) with common spatial and textural characteristics and classification of individual objects using k-nearest neighbors (k-NN) and support vector methods (SVM). Having considered all three approaches to the analysis of bone scintigraphy, it is concluded that the methods based on convolutional neural networks are the most advanced and qualitative. This is confirmed by the growing number of publications on the topic and the general trend of neural networks integration into various spheres of life.

Thus, the aim of the study is to create a convolutional neural network (CNN) for the analysis of 99mTc-MDP SPECT images, capable of using only the frontal part of the image (front view) to determine the presence/absence of foci of pathological accumulation of radiopharmaceutical preparations.

II. DATA COLLECTION AND DATASET PREPARATION

As training data, a set of images and their labels provided by the team of doctors from the Tyumen Medical City Medical Center was used for the study. Each image has a resolution of 1773 × 1037 pixels. An example of an image from the training sample is shown in Figure 2.

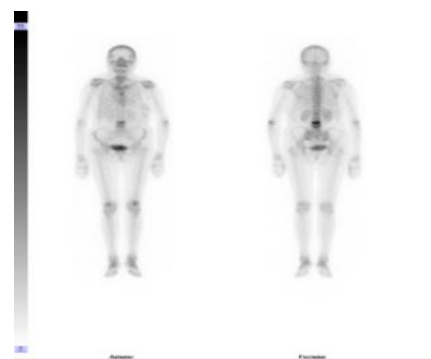


Fig. 2. Examples of images from the dataset

As data preprocessing, the image is cropped. The front view of the person is cropped out of the image. The resolution of the new image is 900×325 . The image obtained after transformation is shown in Figure 3.



Fig. 3. Examples of images after processing

A total of 68 images were provided, 18 of which had foci of pathologic radiopharmaceutical accumulation. An example of human skeletal scintigraphy images with and without pathology is shown in Figure 4.

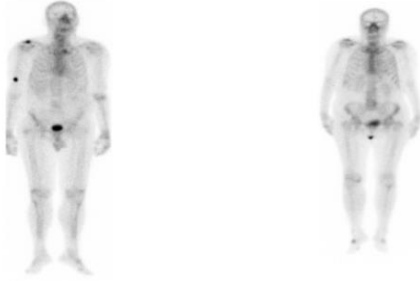


Fig. 4. Examples of images with pathology (left) and without (right)

III. TRAINING OF NEURAL NETWORKS

We offer a solution based on the CNN architecture: InceptionV3, DenseNet121, and 4.3 ResNet101V2, and we will further train them using the scintigraphy images provided to us.

Let us consider the hyperparameters used in training. As a loss function we will take binary_crossentropy. Cross-entropy is widely used in training neural networks, in particular in classification tasks. It measures the difference between the true probability distribution (true labels) and the predicted probability distribution that is obtained at the output of the model. The formula for binary cross-entropy is as follows:

$$L(y_{true}, y_{pred}) = -\frac{1}{N} \sum_{i=1}^N [y_{true,i} \times \ln(y_{pred,i}) + (1 - y_{true,i}) \times \ln(1 - y_{pred,i})],$$

where N is the number of instances in the dataset, $y_{true,i}$ is the true label for the i -th instance, $y_{pred,i}$ is the model prediction for the i -th instance.

The metric used was accuracy, which is calculated as:

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN'}$$

where TP is the number of true positive predictions, TN is the number of true negative predictions, FP is the number of false positive predictions, FN is the number of false negative predictions.

The additional layers used were:

1. Conv2D(32, (1, 1), activation='relu');

2. MaxPooling2D(pool_size=(2, 2));

3. Flatten();

4. Dense(128, activation='relu');

5. Dense(2, activation='softmax').

The last layer, which returns the model prediction, uses the softmax function, which converts the output array into an array of probabilities of belonging to a particular class. The softmax formula: $\delta(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$, where K is the number of classes.

The adam optimizer was used as an optimizer for neural networks; the size of the batch was chosen to be 8, the size of validation data was 0.5, and the data were divided into training and test data in the ratio of 6:4. Neural network architectures were built and trained using the Keras.

IV. EXPERIMENTAL STUDIES

Figures 5-8 show plots of the dependence of the loss function (binary cross-entropy) for all training data and the loss function for validation data on the training epoch. After examining the graphs, we can draw the following conclusions. InceptionV3 architecture on training data has a fairly small loss function, but on validation data the loss function is still large, which may indicate the need to increase the number of epochs or the number of input data.

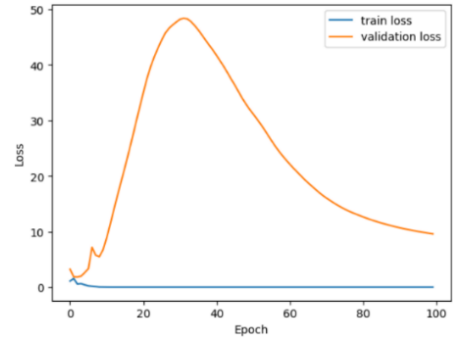


Fig. 5. InceptionV3 Architected Loss Functions

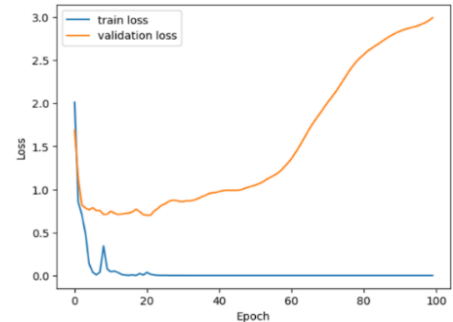


Fig. 6. Loss functions of the DenseNet121 architrave

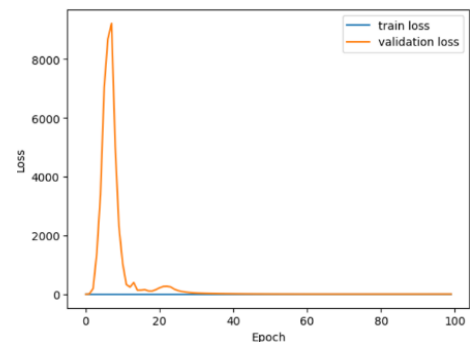


Fig. 7. Loss functions of ResNet101V2 architxture

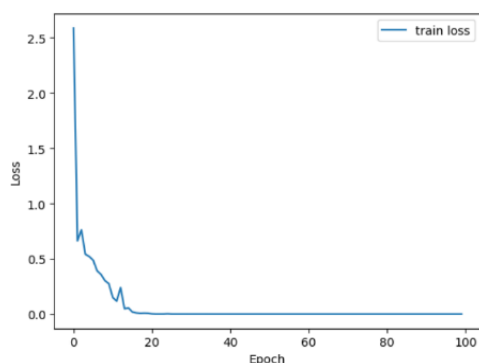


Fig. 8. Loss function on training data of ResNet101V2 architecture

From the graph of the loss function for DenseNet121 we can see that losses for training data fall, but losses for validation data at the end begin to grow, which indicates possible overtraining or incorrectness of the model. For ResNet101V2, the value of the loss function on the validation data rises sharply at the beginning, then falls, and then begins to decrease. While the value of loss function on training data decreases continuously with the increase of training epoch and the model is gradually trained. The comparison of the performance of trained neural networks is presented in Table 1.

TABLE 1. COMPARISON OF TRAINED NEURAL NETWORK PERFORMANCE

Indicator	InceptionV3	DenseNet121	ResNet101V2
Accuracy	0,89	0,93	0,89
Recall	0,33	0,33	0,67
Precision	0,50	1	0,50
Area along the ROC curve	0,65	0,67	0,79

Table I shows that ResNet101V2 has the best Recall score, hence, this architecture was better than the others in finding patients with pathology and not missing them. At the same time the Precision index is the best in DenseNet121, which means that the network has not counted a single healthy patient as a sick one. The area under the ROC curve is the largest in Res-Net101V2. From the table and the above-mentioned we can conclude that out of the three considered CNN architectures ResNet101V2 coped with this task best of all and InceptionV3 worst of all.

V. CONCLUSION

The urgency of developing methods for the automatic localization of foci of pathological accumulation of a radiopharmaceutical drug is due to the need to improve the accuracy of diagnosis and reduce the influence of the human factor in the interpretation of scintigraphic images. The paper examines three main approaches to the analysis of medical images: neural network, morphological and textural. Based on the analysis, the advantage of deep learning methods for solving this problem has been revealed. The novelty of the study lies in the adaptation and comparative analysis of the effectiveness of modern architectures of convolutional neural networks (InceptionV3, DenseNet121 and ResNet101V2) for the task of analyzing bone scintigraphy. The networks were supplemented with five new layers and further trained on a dataset of scintigraphic images provided by specialist doctors. It was found that the ResNet101V2 architecture demonstrated the best performance among the models considered: accuracy was 89%, completeness was 67%, and the area under the ROC curve reached 0.79.

ACKNOWLEDGMENT

The work was carried out within the state assignment of IPISI, NRC “Kurchatov institute”.

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Numerical Simulation of High-frequency Current Flow Processes in Dental Tissues Affected by Caries	184
<i>Ermolaev Nickolaj A., Gorelikova Maria A.</i>	
Investigation of the Chest Model for the Problem of Choosing the Size of Electrode Monitoring Systems for the Regional Ventilation-Perfusion Ratio	188
<i>Gorin Nikita A., Plytkivich Georgii Ye., Shchukin Sergey I.</i>	
Mathematical Modeling and Comparative Control of Direct-Drive Lens System Based on LQR and Neural Network Approach for Pupil Tracking in Teleophthalmology Tasks	192
<i>Mayya Ali Sultan, Moussa Zein Mohammad</i>	
A Physical Phantom of a Premature Baby for Simulation Studies of Thermophysical Processes in Neonatal Incubators	197
<i>Frolov Sergey V., Korobov Artem A., Savinova Kristina S., Potlov Anton Yu., Frolova Tatyana A.</i>	
3D Personalised Modeling of an Aortic Valve	202
<i>Alekseev Denis, Gaisin Shamil, Nazaryan Aikush, Gumerov Ruslan, Afanasyev Ilya, Avsievich Alexandr, Ivaschenko Anton, Kuzmin Andrey</i>	
Development of Device for Objective Monitoring of Valsalva Maneuver	205
<i>Anisimov Aleksei A., Seleznev Dmitry A., Skorobogatova Alena I., Larina Marina A.</i>	
A Hardware and Software Complex for Automated Analysis of the Functional State of Feet Based on Computer Plantograms	209
<i>Mikhailishin Viktor V., Smirnova Ludmila M., Khazov Valentin A.</i>	
A Comparative Evaluation of Spatial Frequency Domain Imaging and Diffuse Reflectance Spectroscopy in Measuring Tissue-Mimicking Phantom Optical Properties	213
<i>Kolpakov Alexander V., Sokolova Diana Yu., Semenova Anastasia S.</i>	
Electrical Impedance Mapping Method of Back Muscles in the Area of Damaged Intervertebral Discs	218
<i>Blinov Andrei A.</i>	

Section 3. Medical Decision Support Systems. Intelligent Systems in Healthcare

Fuzzy Models for Assessing the Protective Functions of the Body Based on Its Antioxidant Status for Medical Expert Systems	223
<i>Korenevsky Nikolay A., Rodionova Sofya N., Seregin Stanislav P., Kholimenko Ivan M., Tsukanova Margarita N.</i>	
Growth Prediction of Primary Extra-Axial Brain Tumors of Magnetic Resonance Images Based on Deep Learning	229
<i>Selezneva Alisa A., Ilyasova Natalya Yu., Demin Nikita S., Surovcev Evgeniy N., Kapishnikov Aleksandr V.</i>	
Accuracy and Efficiency of Neural Network Models in the Diagnosis of Obstructive Micturition	233
<i>Panferov Alexandr S., Gadzhiev Nariman K., Yastrebov Vitaly S., Filist Sergey A., Puchenkov Kirill I., Zhilin Ilya A.</i>	
sEMG-based Hand Gesture Recognition Using CNN	238
<i>Assalama Lara, Potekhin Vyacheslav V.</i>	
Development of a Clinical Decision Support System on Selection of Laser Radiation Characteristics and Component Composition of Laser Solder for Laser Reconstruction of Biological Tissue Integrity	242
<i>Suchkova Victoria V., Ryabkin Dmitry I., Selishchev Sergey V., Gerasimenko Alexander Yu.</i>	
Extraction of Informative Features for Optical Biopsy Data Analysis	245
<i>Matveeva Irina</i>	
Comparison of Neural Network and Hybrid Approaches in Analyzing Spine Radiographs of Osteoporosis Diagnosis	249
<i>Nekrasova D.V., Zarov E.V., Ilyasova Natalya Yu., Demin Nikita S., Pervushkin S.S.</i>	
A Novel Deep Learning Prototype for Alzheimer's Disease Detection	253
<i>Karim Sabry Ramadan Attia, Housam Hasan Bou Issa, Alrawi Adnan</i>	
Module for Automated Assessment of Visual Function Safety in the Diagnosis of Multiple Sclerosis	256
<i>Bolsunov Konstantin N., Gapanenok Alexandra E.</i>	
Technology of Localization Pathological Accumulation Foci of Radiopharmaceutical Based on Deep Learning Methods	260
<i>Ilyasova Natalya Yu., Gulinskaya Irina I., Demin Nikita S., Delov Konstantin E., Alekhin Eduard N.</i>	